Standard structures have the standard numerals as their domain

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openlogic:model-theory/models-of-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:standard-domain of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the definition of a standard structure for the language of arithmetic
- isomorphisms of L_A-structures
- the interpretation of numerals in a structure

Proposition 0.1. *If !!astructure $\mathfrak{A}M$ is standard, then its domain is the set of values of the standard numerals, i.e.,*

$$|\mathfrak{A}|M = \{\, \mathrm{val}(\underline{n})M \mid n \in \mathbb{N} \,\}$$

Proof. Clearly, every $\mathrm{val}(\underline{n})M \in |\mathfrak{A}|M$. We just have to show that every $x \in |\mathfrak{A}|M$ is equal to $\mathrm{val}(\underline{n})M$ for some n. Since $\mathfrak{A}M$ is standard, it is isomorphic to $\mathfrak{A}N$. Suppose $g\colon \mathbb{N} \to |\mathfrak{A}|M$ is an isomorphism. Then $g(n) = g(\mathrm{val}(\underline{n})N) = \mathrm{val}(\underline{n})M$. But for every $x \in |\mathfrak{A}|M$, there is an $n \in \mathbb{N}$ such that $g(n) = x$, since g is !!surjective. □

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